Time-Varying Wideband Interference Mitigation for SAR via Time-Frequency-Pulse Joint Decomposition Algorithm

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***Abstract*— Wideband interference (WBI) may severely affect the imaging quality of synthetic aperture radar (SAR) systems. Since it highly overlaps with useful signals in the 1-dimensional (1-D) time or frequency domain, the existing WBI mitigation methods usually transform 1-D echoes into a 2-D transform domain. However, they usually suffer from a model mismatch, which results in the loss of the useful signal. To tackle this problem, a novel algorithm combining time-frequency-pulse (TFP) joint characteristics and robust principal component analysis (RPCA) is proposed for WBI mitigation. The TFP joint feature of SAR echo is introduced for interference mitigation for the first time. We first transform the SAR echoes into the time–frequency domain and construct a new TFP matrix by reshaping the STFT matrices between adjacent pulses. In terms of WBI-occupied SAR echoes, the short-time Fourier transformation (STFT) in adjacent pulses can be modeled as a combination of a low-rank part (i.e., useful SAR echoes) and a sparse counterpart (i.e., WBIs), which well fits the assumption of RPCA. Then, the TFP matrix is decomposed into the useful signal TFP matrix and the WBI TFP part by taking full advantage of the low-rank and sparse properties. Finally, the WBIs can be reconstructed and subtracted from the echoes to realize interference mitigation. Experimental results on both simulated and measured datasets show that the proposed algorithm not only suppresses WBIs effectively but also preserves useful information as much as possible.**

***Index Terms*— Robust principal component analysis (RPCA), synthetic aperture radar (SAR), time-frequency-pulse (TFP) domain, wideband interference (WBI).**

I. Introduction

DUE to its ability to operate day and night under all weather conditions, synthetic aperture radar (SAR) systems have been extensively used in civil and scientific exploration applications. However, the rapid development of electronic instruments sharing the common spectrum as the SAR systems poses a huge threat to accurate remote sensing with SAR systems [1], [2], [3]. According to its bandwidth, the interference can be classified into narrowband interference (NBI) and wideband interference (WBI). Compared with NBI, WBI occupies a larger bandwidth of useful signals, which makes it more difficult to mitigate. Therefore, WBI detection and mitigation approaches have drawn increasing attention in the field of SAR systems [4].

In recent years, plenty of methods were proposed for WBI mitigation, and these methods can be classified into three categories: parametric, nonparametric, and semiparametric.

1) Parametric Methods: For these kinds of WBI mitigation methods, the main idea is to model the WBIs as a summation of higher order modulated signals, whose parameters, such as amplitude, frequency, second-order phase, and high-order phase, can be estimated by the fractional Fourier transform (FRFT) [4], high-order ambiguity function (HAF) [5], product HAF (PHAF) [6], polynomial phase transform (PPT) [7], and iterative adaptive approach (IAA) [8]. Then, the time-varying WBI can be iteratively reconstructed by estimated parameters and mitigated by subtracting from the WBI-contaminated SAR echoes. However, when prior knowledge is absent, model mismatch and parameter estimation error are problematic leading to the degradation of signal synthesis. In addition, they are not time-efficient because of the high-dimensional parameters search involved.

2) Nonparametric Methods: Unlike parametric methods, nonparametric methods do not require exact WBI signal modeling and multiparameter estimation. Generally, these methods mainly include three steps. First, 1-dimensional (1-D) SAR echoes are transformed into the 2-D transform domain. In practice, short-time Fourier transform (STFT) [9], [10], [11], [12], Wigner–Ville distribution (WVD) [13], [14], ambiguity function (AF) [15], and wavelet transform (WT) [16] are mainly selected as 2-D transform domains since their inverse transform can be easily implemented. Then, property differences in the transform domain are exploited to design 2-D notch filters or construct the subspace projection to achieve interference mitigation. Finally, the WBI-free SAR echoes are reconstructed by inverse 2-D transformations. The main drawback of these methods that limit their applications is the insufficient exploitation of the WBI characteristics, resulting in the loss of the useful signal when they are overlapping with the WBIs to a large extent.

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3) Semiparametric Methods: They were used for interference mitigation by solving hyperparameter optimization problems [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27]. The core idea of these methods is to separate or mitigate the interference by making full use of the low-rank and sparse properties of the SAR echoes contaminated by the interference. Compared with parametric methods, these methods reduce the difficulty and complexity of multiparameter estimation. Tensor techniques for interference mitigation are proposed with good performance for complex interferences [25], [26]. However, they are suitable for interferences with stable frequencies in multiple views. The frequency variation of interference would break the constraints of low-rank and sparse properties and result in performance degradation [11].

In this article, a robust principal component analysis (RPCA)-based time-frequency-pulse (TFP) joint decomposition algorithm (RPCA-TFP-JDA) is proposed. This algorithm makes full use of the TFP joint characteristics of WBI and useful signals. For WBI-contaminated SAR echoes, the TF characteristics of useful signals between the adjacent pulses are almost identical, which satisfies the low-rank property. Meanwhile, the time-varying WBI is sparsely distributed and their TF locations change dramatically in adjacent pulses, and thus, it can be considered as a sparse part. According to these TFP properties, an RPCA approach is utilized to decompose a low-rank part (i.e., useful SAR echoes) and a sparse counterpart (i.e., WBIs).

The main contributions of this article can be summarized as follows.

1) A log t-constant false alarm rate (CFAR) detector is introduced for WBI identification in the TF domain. The TF detector is utilized to determine whether interference exists or not at each slice of the TF domain. Meanwhile, the statistical histogram of the TF amplitude spectrogram of SAR echo signal is analyzed, and the result of airborne measured SAR data shows that it approximately obeys the Weibull distribution. Hence, a log t-CFAR detector is applied as it is suitable for detection in the Weibull background.
2) The TFP joint characteristics are introduced for WBI mitigation for the first time. The low-rank and sparse properties of useful signals and WBIs in the TFP domain are demonstrated in this article. Furthermore, as a semiparametric method, we propose an adaptive parameter estimation method for determining the sparsity of WBIs, which is a crucial parameter for the proposed method.
3) The WBI mitigation experiments on both simulated and measured datasets are applied, and the proposed algorithm achieves better WBI mitigation performance than the existing algorithms, even when the useful signal and WBIs are highly overlapped in the TF domain.

The rest of this article is organized as follows. In Section II, signal models for WBI-polluted SAR echo and formulation of STFT are introduced. In Section III, the TF interference detector and RPCA-TFP-JDA are presented in detail. The experimental results and performance analysis are shown in Section IV, and Section V provides the conclusion.

II. Related Work

A. Signal Model

In a complex electromagnetic environment, the SAR system may be severely interfered by other radio services operating in the same frequency bands. Assume that a single-channel SAR system transmits Q pulses, and each received signal consists of N range samples; then, the SAR received signal in the qth pulse (i.e., slow time sample or azimuth sample) can be formulated as

$$x_q(n) = U_q(n) + I_q(n) + \text{Noise}_q(n), \quad q = 1, \ldots, Q \tag{1}$$

where $U_q(n)$, $I_q(n)$, and $\text{Noise}_q(n)$ represent the scene reflected signal (i.e., useful signal), interferences, and additive noise in the qth pulse, respectively, and $n = 1, 2, \ldots, N$ denotes the sample sequence of the fast time.

WBIs usually occupy most of the frequency bands of useful signals, affecting the quality of the SAR images severely. Generally, two typical forms of WBI are considered, i.e., chirp-modulated (CM)-WBI and sinusoidal-modulated (SM)-WBI [26]. Specifically, the CM-WBI can be defined as

$$\text{WBI}_{\text{CM},q}(n) = \sum_{l=1}^{L} A_{q,l} \exp\left(j2\pi f_{q,l} n + j\pi \kappa_{q,l} n^2\right) \tag{2}$$

where $A_{q,l}$ denotes the amplitude and $f_{q,l}$ and $\kappa_{q,l}$ represent the frequency and chirp rate of the lth CM-WBI in the qth pulse, respectively. The SM-WBI can be formulated as

$$\text{WBI}_{\text{SM},q}(n) = \sum_{l=1}^{L} A_{q,l} \exp\left\{ j\beta_{q,l} \sin\left(2\pi f_{q,l} n + \varphi_{q,l}\right)\right\} \tag{3}$$

where $\beta_{q,l}$ and $\varphi_{q,l}$ denote the modulation index and initial phase of the lth SM WBI in the qth pulse, respectively.

B. STFT

For a received signal, WBIs highly overlap with useful signals in the time and frequency domains, which makes it difficult to separate the useful signal from WBI using 1-D time or frequency methods. To better eliminate WBIs, previous researchers transform the 1-D received signal $x_q(n)$ into 2-D TF joint domain using STFT, which is defined as

$$\text{STFT}_q(n, k) = \sum_{m=0}^{M-1} h(m) x_q(n+m) e^{-j2\pi mk/M} \tag{4}$$

where $h(m)$ is the window function and its width is set to M. In (4), $\text{STFT}_q(n, k)$ denotes an STFT value of the qth pulse with a given time n and discrete frequency k. Its vector form can be written as

$$\mathbf{STFT_{q,M}}(n) = \mathbf{W_M H_M x_q}(n) \tag{5}$$

where $\mathbf{H_M}$ is a diagonal $M \times M$ matrix and the values on the diagonal denote as $H(m, m) = h(m)$, $m = 0, 1, 2, \ldots, M-1$; and $\mathbf{STFT_{q,M}}(n)$ and $\mathbf{x_q}(n)$ are vectors

$$\mathbf{STFT_{q,M}}(n) = \left[\text{STFT}_q(n, 0), \text{STFT}_q(n, 1) \ldots, \text{STFT}_q(n, M-1)\right]^{\text{T}} \tag{6}$$

$$\mathbf{x_q}(n) = \left[x_q(n), x_q(n+1), \ldots, x_q(n+M-1)\right]^{\text{T}} \tag{7}$$

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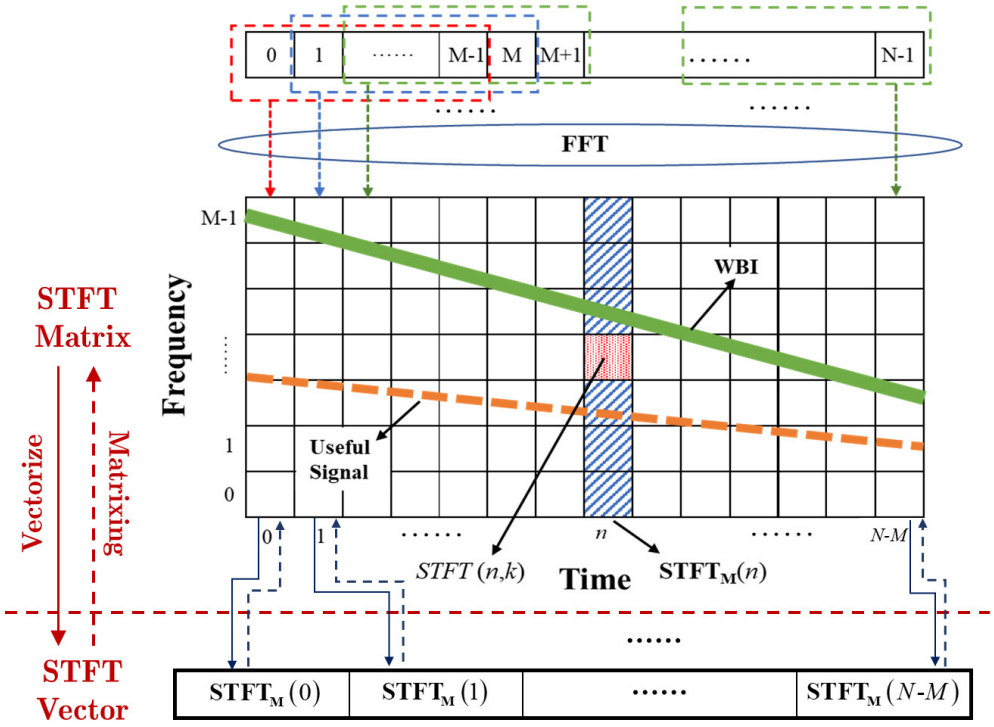


Fig. 1. Construction of STFT matrix and STFT vector.

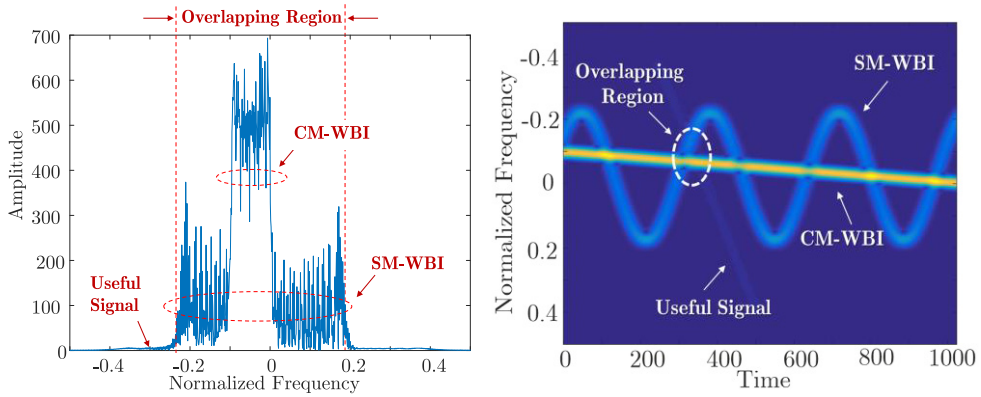


Fig. 2. Illustrations of useful signal with CM-WBI and SM-WBI in (a) 1-D frequency domain and (b) 2-D TF domain.

and $\mathbf{W_M}$ represents the $M \times M$ coefficient matrix of discrete Fourier transform (DFT).

The STFT matrix can be obtained by stacking the STFT vectors as the columns of the STFT matrix, as shown in Fig. 1, denoted by

$$\mathbf{STFT_q} = \left[\mathbf{STFT_{q,M}}(0)\ \mathbf{STFT_{q,M}}(1)\ \cdots\ \mathbf{STFT_{q,M}}(N-M)\right]. \tag{8}$$

The STFT matrix of the received signal in the qth pulse can be written as follows:

$$\mathbf{STFT_{X,q}} = \mathbf{STFT_{U,q}} + \mathbf{STFT_{I,q}} + \mathbf{STFT_{N,q}} \tag{9}$$

where $\mathbf{STFT_{U,q}}$, $\mathbf{STFT_{I,q}}$, and $\mathbf{STFT_{N,q}}$ denote the STFT matrices of the useful signal, interference, and noise, respectively. Fig. 2 shows the useful signal contaminated by CM-WBI and SM-WBI in the 1-D frequency domain and 2-D TF domain, respectively. It can be seen from Fig. 2(a) that the CM-WBI and SM-WBI are highly overlapped with useful signals in the frequency domain, while in the 2-D TF domain, they are highly concentrated with much smaller overlapping regions, as shown in Fig. 2(b). However, both in the frequency domain and TF domain, the useful signals may also be eliminated by filtering methods, when the WBIs and useful signal are in the same region.

III. WBI Mitigation Scheme via RPCA-TFP-JDA

In this section, the framework and main procedures of the WBI mitigation scheme are introduced first. Then, the interference detection in the TF domain based on the log t-CFAR detector is presented. After interference identification,

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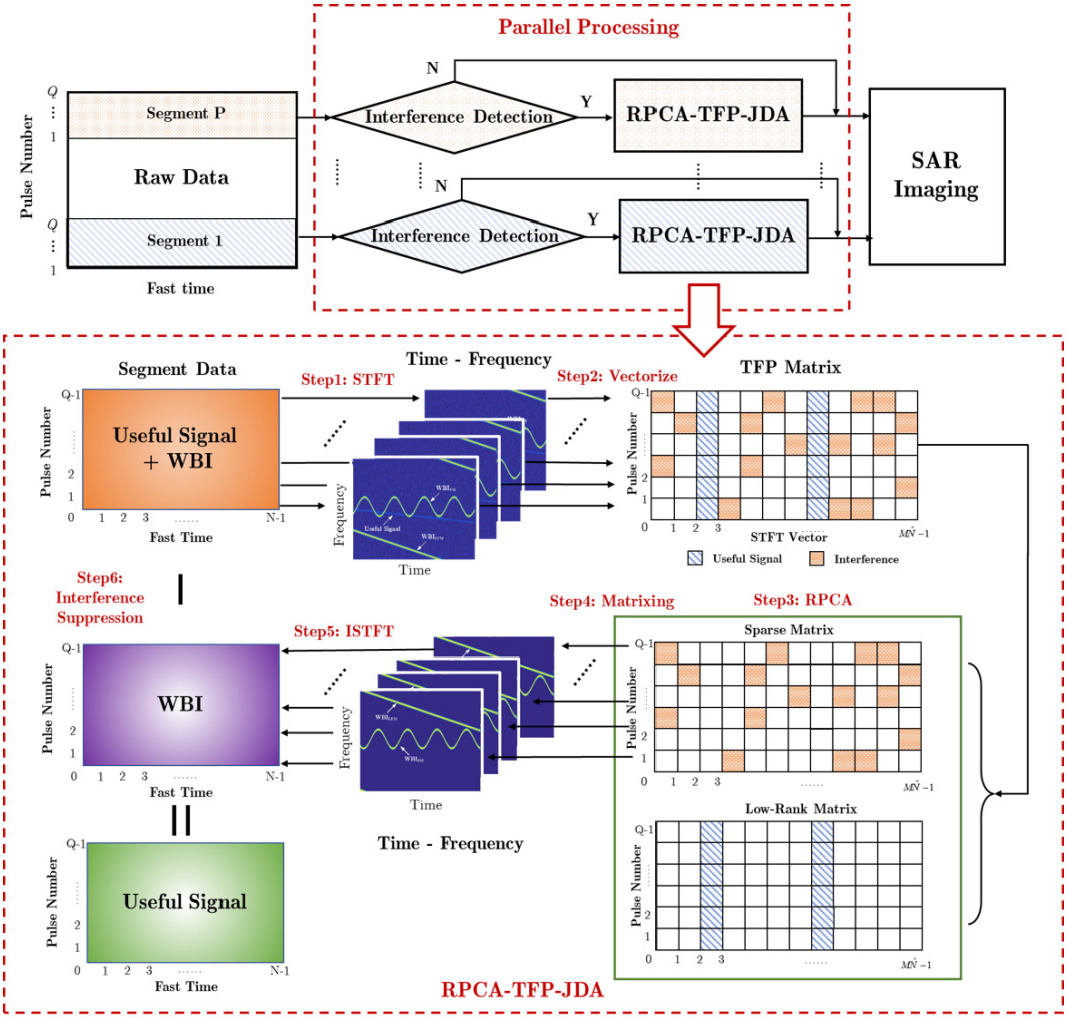


Fig. 3. Flowchart of interference mitigation based on RPCA-TFP-JDA.

RPCA-TFP-JDA is proposed to separate the WBIs from the useful signal.

A. Flowchart of the Proposed Algorithm

The flowchart of the proposed algorithm is presented in Fig. 3, and the raw data are divided into P segments for parallel processing, each of which contains Q pulses. First, the interference-contaminated SAR echo is detected adaptively in the TF domain via the log t-CFAR detector. Then, RPCA-TFP-JDA is utilized to ensure that WBIs in each segment are eliminated. Finally, a well-focused SAR image is obtained by the conventional radar imaging algorithm.

In the following, the proposed algorithm, including the TF interference detector and RPCA-TFP-JDA, is discussed in detail.

B. TF Interference Detector

For effective WBI mitigation, it is of great significance to detect where interference exists or not. If interference exists, the number of points that WBI occupies in the TFP domain (i.e., sparsity) needs to be estimated as a critical parameter in RPCA-TFP-JDA. Here, the TF detector is utilized to realize interference detection and sparsity estimation. The core idea is to detect interference at each time slice in the TF domain. Generally, the interference detector can be mainly modeled by a binary hypothesis test, i.e., H_1 for interference present and H_0 for interference absent. The key issue is to determine an adaptive detection threshold, which is usually related to the TF amplitude distribution of SAR echoes. Fig. 4 shows the TF spectrogram and TF amplitude distribution of measured RFI-free SAR data. Based on the parameter estimation [28] and goodness-of-fit test [29], we analyze the statistical characteristics of the TF amplitude spectrum of measured data collected by the airborne SAR system, three distribution models, i.e., Rayleigh, Log-normal, and Weibull distribution, are applied to fit the TF amplitude statistical histogram, as shown in Fig. 4(b). The mean error values of these three distributions are calculated and the result of the Weibull distribution reaches the minimum, indicating that the TF amplitude spectrogram approximately obeys the Weibull distribution.

Since it is suitable for Weibull distribution, the log t-CFAR detector is applied with its threshold being given as follows [30]:

$$\log\left(|\mathrm{STFT}_q(n,k)|\right) \underset{H_0}{\overset{H_1}{\gtrless}} \mathrm{T}_h$$
$$\text{where} \quad \mathrm{T}_h = \max\{\mu_r + \alpha\hat{\sigma}_r, \mu_c + \alpha\hat{\sigma}_c\} \tag{10}$$

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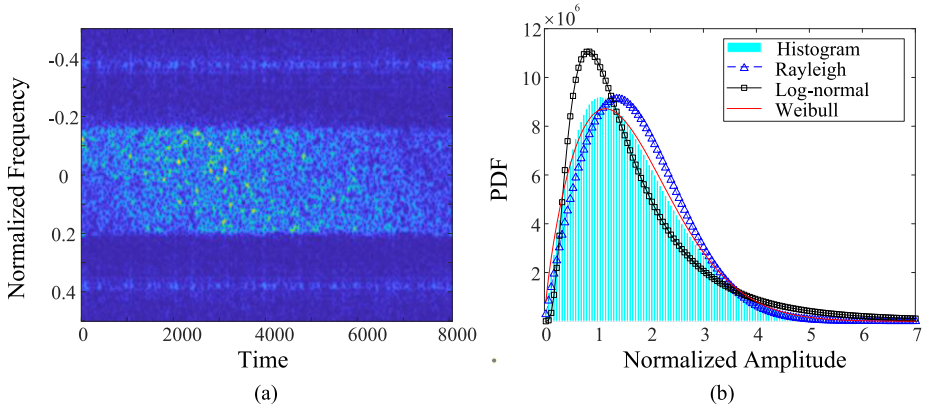


Fig. 4. (a) TF spectrogram and (b) TF amplitude distribution of measured RFI-free SAR data.

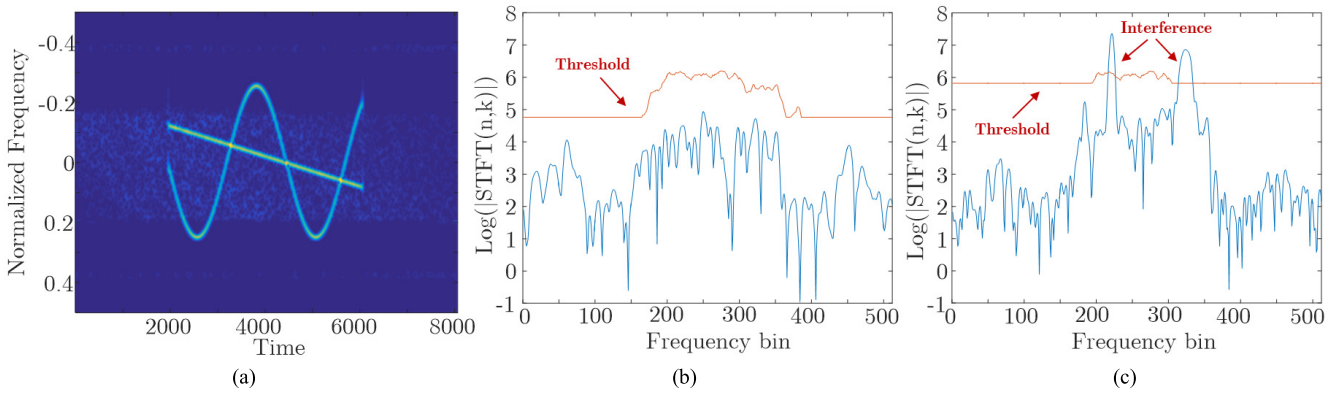


Fig. 5. (a) TF spectrogram with WBI and log TF amplitude spectrogram at (b) 1000th and (c) 3000th time slices, respectively.

where $\mathrm{STFT}_q(n,k)$ is the magnitude of the cell under test (CUT), μ_r and $\hat{\sigma}_r$ are the estimated mean and standard deviation of the kth row of the STFT matrix, respectively, μ_c and $\hat{\sigma}_c$ are the estimated mean and standard deviation of the nth column of the STFT matrix, respectively, and α is the threshold multiplier and its range is set to [1.5, 2.0] empirically. The TF spectrogram of measured data with simulated CM-WBI and SM-WBI is given in Fig. 5(a), and the log TF amplitudes' spectrogram at the 1000th and the 3000th time slices is shown in Fig. 5(b) and (c), respectively. The threshold is calculated by (10) with $\alpha = 1.8$. From Fig. 5(b) and (c), it is obvious that the threshold can be adjusted adaptively in the interference and noninterference cases.

C. RPCA-TFP-JDA

To preserve useful signals well, the pulse dimension is exploited to figure out the intrinsic difference between the interferences and useful signals. For a better description, three pulses of STFT matrices are given as an example shown in Fig. 6.

From the figures, two remarks can be summarized as follows.

Remark I: The TF spectra of useful signals in adjacent pulses are located in the same time and frequency bins, and thus, we can conclude that useful signals in the TFP domain satisfy the low-rank property.

Remark II: The TF spectra of WBIs are sparsely distributed in adjacent pulses with different locations, and thus, CM-WBI and SM-WBI have sparse characteristics.

For detailed discussion and proof, please refer to Appendix A.

The main problem is how to separate the interferences and useful signals using the low-rank and sparse characteristics. Fortunately, RPCA provides an excellent solution to settle this problem. The basic model of RPCA is

$$\mathbf{X} = \mathbf{L} + \mathbf{S} + \mathbf{N} \tag{11}$$

where $\mathbf{L}$, $\mathbf{S}$, and $\mathbf{N}$ represent a low-rank matrix, a sparse matrix, and a noise matrix, respectively. Assume that the rank of matrix $\mathbf{L}$ is low and nonzero elements in matrix $\mathbf{S}$ are sparse, and the approximated "low-rank + sparse" matrix $\mathbf{X}$ can be decomposed into a low-rank matrix and a sparse matrix using the Go Decomposition (GoDec) algorithm [18], summarizing as an optimization problem

$$\begin{cases} \min\limits_{\mathbf{L},\mathbf{S}} & \|\mathbf{X}-\mathbf{L}-\mathbf{S}\|_F^2 \\ \text{s.t.} & \mathrm{rank}(\mathbf{L}) \le r \\ & \mathrm{card}(\mathbf{S}) \le \varepsilon \end{cases} \tag{12}$$

where $||\cdot||_F$ denotes the Frobenius norm, rank$(\cdot)$ denotes the rank operator, card$(\cdot)$ denotes the cardinality of the sparse matrix, r represents the rank of $\mathbf{L}$, and ε represents the sparse degree of $\mathbf{S}$. The GoDec algorithm alternatively assigns the r-rank approximation of $\mathbf{X}$–$\mathbf{S}$ to $\mathbf{L}$

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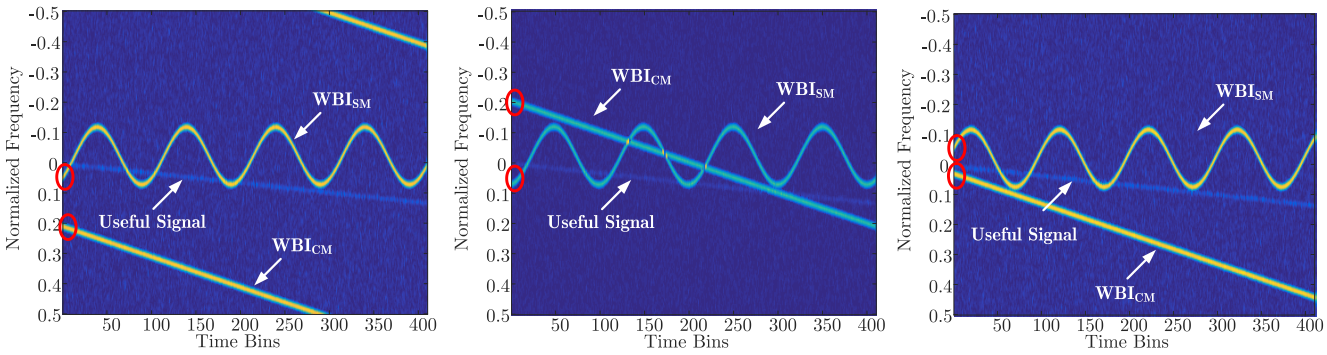


Fig. 6. TF spectra in (a) $(q-1)$th pulse, (b) qth pulse, and (c) $(q+1)$th pulse.

and assigns the sparse approximation with cardinality ε of **X–L** to **S**. The updating of **L** is obtained via singular value hard thresholding of **X–S**, while the updating of **S** is obtained via entrywise hard thresholding of **X–L**. The bilateral random projection (BRP)-based low-rank approximation is applied to accelerate the r-rank of the approximation of **X–S**. More details of the GoDec algorithm can refer to [18].

To well fit the RPCA expression, we first vectorize each pulse's STFT matrix (shown in Fig. 1) and then stack each vector as one of the rows in a new matrix $\mathbf{G_X}$ (denoted by the TFP matrix), which can be written as (13), shown at the bottom of the next page.

The TFP matrix is composed of three parts, i.e., a low-rank part of useful signal $\mathbf{G_L}$, a sparse part of WBIs $\mathbf{G_S}$, and a noise matrix $\mathbf{G_N}$, which are expressed as

$$\mathbf{G_X} = \mathbf{G_L} + \mathbf{G_S} + \mathbf{G_N}. \tag{14}$$

According to (A9), the TFP matrix of useful signal $\mathbf{G_L}$ can be rewritten as follows: (15), shown at the bottom of the next page.

From (15), it is noted that $\mathbf{G_L}$ is composed of a $Q \times 1$ column vector multiplied by a $1 \times M\hat{N}$ row vector, where $\hat{N} = N - M + 1$, which proves that the TFP matrix of useful signal $\mathbf{G_L}$ is a one-rank matrix. Therefore, the parameter rank "r" in RPCA is set to 1, while, according to (A11), the TFP matrix of WBI $\mathbf{G_S}$ cannot be expressed as a column vector multiplied by a row vector. Therefore, the sparse TFP matrix of WBI and the low-rank TFP matrix of the useful signals can be separated by the RPCA method. According to RPCA, there are two methods to obtain the useful signal.

Method I: Reconstruct the WBIs from the sparse TFP matrix and subtracted them from the original signal to obtain the useful signal.

Method II: Reconstruct the useful signal directly from the low-rank TFP matrix.

The comparative experimental results show that the signal obtained by Method I is much closer to the useful signal than that obtained by Method II. Thus, we adopt Method I in this article.

The flowchart of WBI mitigation using the RPCA-TFP-JDA is illustrated in the dotted rectangle, as shown in Fig. 3, and the detailed processing steps of our scheme can be summarized as the following steps.

Step 1: Extract the qth pulse of the received signal $x_q(n)$, where $1 \le n \le N$ is the sampling number and $1 \le q \le Q$ is the pulse number, and then calculate its STFT matrix with a size of $M \times \hat{N}$.

Step 2: Vectorize the STFT matrix into a $1 \times M\hat{N}$ vector as one of the rows in the TFP matrix. Let $q = q+1$, and if q is less than or equal to Q, iterate the above steps; then, we can obtain the TFP matrix $\mathbf{G_X}$ of size $Q \times M\hat{N}$.

Step 3: According to Section III-C, the formed matrix $\mathbf{G_X}$ can be decomposed into a low-rank TFP matrix of useful signal $\mathbf{G_L}$ and a sparse TFP matrix of WBI $\mathbf{G_S}$ by the GoDec algorithm.

Step 4: Extract the qth row of the sparse TFP matrix $\mathbf{G_S}$ and then transform it into an $M \times \hat{N}$ STFT matrix.

Step 5: Reconstruct the WBIs from the STFT matrix via inverse STFT (ISTFT) operation.

Step 6: Subtract the reconstructed WBIs from the received signal in the qth pulse; then, let $q = q+1$ and iterate Steps 4–6 until q equals Q. Then, the WBIs in each pulse are eliminated. Finally, a well-focused SAR image can be obtained by the conventional radar imaging algorithm.

IV. Experimental Results

In this section, the effectiveness of the proposed scheme is demonstrated with both simulated and real SAR data.

A. Result of Simulated Data

To demonstrate the validity of the proposed method, the simulated three-point-target raw data contaminated by CM-WBI and SM-WBI is generated. The parameters of the radar system are listed in Table I.

The radar echo can be modeled by

$$x_q(n) = U_q(n) + I_{q,\mathrm{CM}}(n_q) + I_{q,\mathrm{SM}}(n_q) \tag{16}$$

where

$$U_q(n) = \sum_{i=1}^{3} \exp\left[j\pi \times 6 \times 10^{12} \times \left(n - \frac{2R_{i,q}}{c}\right)\right] \times \exp\left[-j4\pi \times 10^9/c \times R_{i,q}\right] \tag{17}$$

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TABLE I
PARAMETERS OF RADAR SYSTEM

Parameter	Value
Carrier Frequency (f_c)	1 *GHz*
Bandwidth of Transmitted Signal (B)	30 *MHz*
Time width of Transmitted Signal (T_p)	5 *us*
Pulse Repetition Time (*PRT*)	30 *ms*
Altitude of Platform (H)	5 *km*
Velocity of Platform (V)	100 *m/s*
Sampling Frequency (f_s)	90 *MHz*

$$I_{q,\mathrm{CM}}(n_q) = 10 \times \exp\Big[j2\pi \times 10^6 \times (n_q/f_s) + j\pi \times 10^{12} \times (n_q/f_s)^2\Big] \quad (18)$$

$$I_{q,\mathrm{SM}}(n_q) = 8 \times \exp\big\{j2\pi \times 10^6 \times (n_q/f_s) + j\pi \times 15 \times \cos\big[\pi \times 10^5 \times (n_q/f_s)\big]\big\} \quad (19)$$

where $U_q(n)$, $I_{q,\mathrm{CM}}(n_q)$, and $I_{q,\mathrm{SM}}(n_q)$ represent the simulated point-target echoes, CM-WBI, and SM-WBI of the qth pulse, respectively; n is a fast sampling number; and $n_q = (q-1) \times \mathrm{PRT} \times f_s + n$. c is the speed of light, and $R_{i,q}$ is the slope distance of the ith point target from the platform at the qth pulse, which is defined by

$$R_{i,q} = \sqrt{(V \times (q-1) \times \mathrm{PRT} - x_{i,P})^2 + y_{i,P}^2 + (H - z_{i,P})^2} \quad (20)$$

where $x_{i,P}$, $y_{i,P}$, and $z_{i,P}$ denote the coordinate of the ith point target, V represents the velocity of the platform, and H is the height of the platform from the ground. The geometric configuration of a side-looking SAR system is shown in Fig. 17. In this simulation, the coordinates of the three point-target are Target A (0, 10000, 0), Target B (50, 10100, 0), and Target C (100, 10250, 0).

Fig. 7 shows the comparison of WBI mitigation among frequency-notch filter [31], TF-notch filter [9], and the proposed method in adjacent three pulses. The left, middle, and right columns of subfigures show the TF spectra obtained by three interference mitigation methods in the 1st to the 3rd pulse, respectively. The Hamming window of length 128 is applied. Fig. 7(a) shows the TF spectra of the received signals. It is shown that the TF spectra of useful signals are located at the same time and frequency bins in adjacent pulses, while TF locations of CM-WBI and SM-WBI are different, which is consistent with the analysis in Section III-C. For comparison, the results of the frequency-notch filter and TF-notch filter are provided. Fig. 7(b) shows the results of WBI eliminated by the frequency-notch filter. It is obvious that the useful signals, whose spectra overlap with WBIs, are filtered out by the frequency-notch filter method. Fig. 7(c) shows the TF spectra obtained by the TF-notch filter method. Different from the frequency-notch filter, the TF-notch filter method transforms the received signal into the TF domain and mitigates the WBIs by designing a 2-D filter. It should be noted that only a fraction of WBIs is located in the same regions with useful signals, which implies that the TF-notch filter method can mitigate interferences with much smaller energy loss of useful signals than the frequency-notch filter does. Fig. 7(d) shows the results obtained by the proposed method with $r = 1$ and $\varepsilon = 1.2 \times 10^6$. It is obvious from Fig. 7(d) that the TF spectra of useful signals are preserved well, even when the useful signals are overlapped with WBIs in the TF domain.

To make a quantitative evaluation for the proposed algorithm, a signal distortion ratio (SDR) is introduced, which is defined by [16]

$$\mathrm{SDR} = 10\log_{10}\left(\frac{\sum_q \sum_n |d_0(n,q) - \hat{d}(n,q)|^2}{\sum_q \sum_n |d_0(n,q)|^2}\right) \quad (21)$$

where $\hat{d}(n,q)$ represents the signal after interference mitigation and $d_0(n,q)$ denotes the original signal without interference. The SDRs of these three WBI mitigation methods are calculated and their results are listed in Table II. In Table II, the SDR of the proposed method is much lower than that of the other two methods, indicating that the proposed method has sound performance over the frequency-notch filter and TF-notch filter methods.

Fig. 8 shows the comparison of simulated point-target imaging results with and without WBIs. Comparing Fig. 8(a) with Fig. 8(b), it is obvious that interferences degrade the imaging quality severely. Fig. 9 shows the point-target imaging

$$\mathbf{G_X} = \begin{bmatrix} \mathbf{STFT}_{1,\mathbb{M}}(0)^{\mathrm{T}} & \mathbf{STFT}_{1,\mathbb{M}}(1)^{\mathrm{T}} & \cdots & \mathbf{STFT}_{1,\mathbf{M}}(N-1)^{\mathrm{T}} \\ \mathbf{STFT}_{2,\mathbb{M}}(0)^{\mathrm{T}} & \mathbf{STFT}_{2,\mathbb{M}}(1)^{\mathrm{T}} & \cdots & \mathbf{STFT}_{2,\mathbf{M}}(N-1)^{\mathrm{T}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{STFT}_{\mathbf{Q},\mathbf{M}}(0)^{\mathrm{T}} & \mathbf{STFT}_{\mathrm{Q},\mathrm{M}}(1)^{\mathrm{T}} & \cdots & \mathbf{STFT}_{\mathrm{Q},\mathrm{M}}(N-1)^{\mathrm{T}} \end{bmatrix} \begin{matrix} \text{1-th pulse} \\ \text{2-th pulse} \\ \vdots \\ \text{Q-th pulse} \end{matrix} \quad (13)$$

$$\mathbf{G_L} \approx \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{Q\times 1} \begin{bmatrix} \mathbf{STFT}_{\mathbf{1},\mathbf{M}}(0)^{\mathrm{T}} & \mathbf{STFT}_{\mathbf{1},\mathbf{M}}(1)^{\mathrm{T}} & \cdots & \mathbf{STFT}_{\mathbf{1},\mathbf{M}}(N-M)^{\mathrm{T}} \end{bmatrix}_{1\times M\hat{N}} \quad (15)$$

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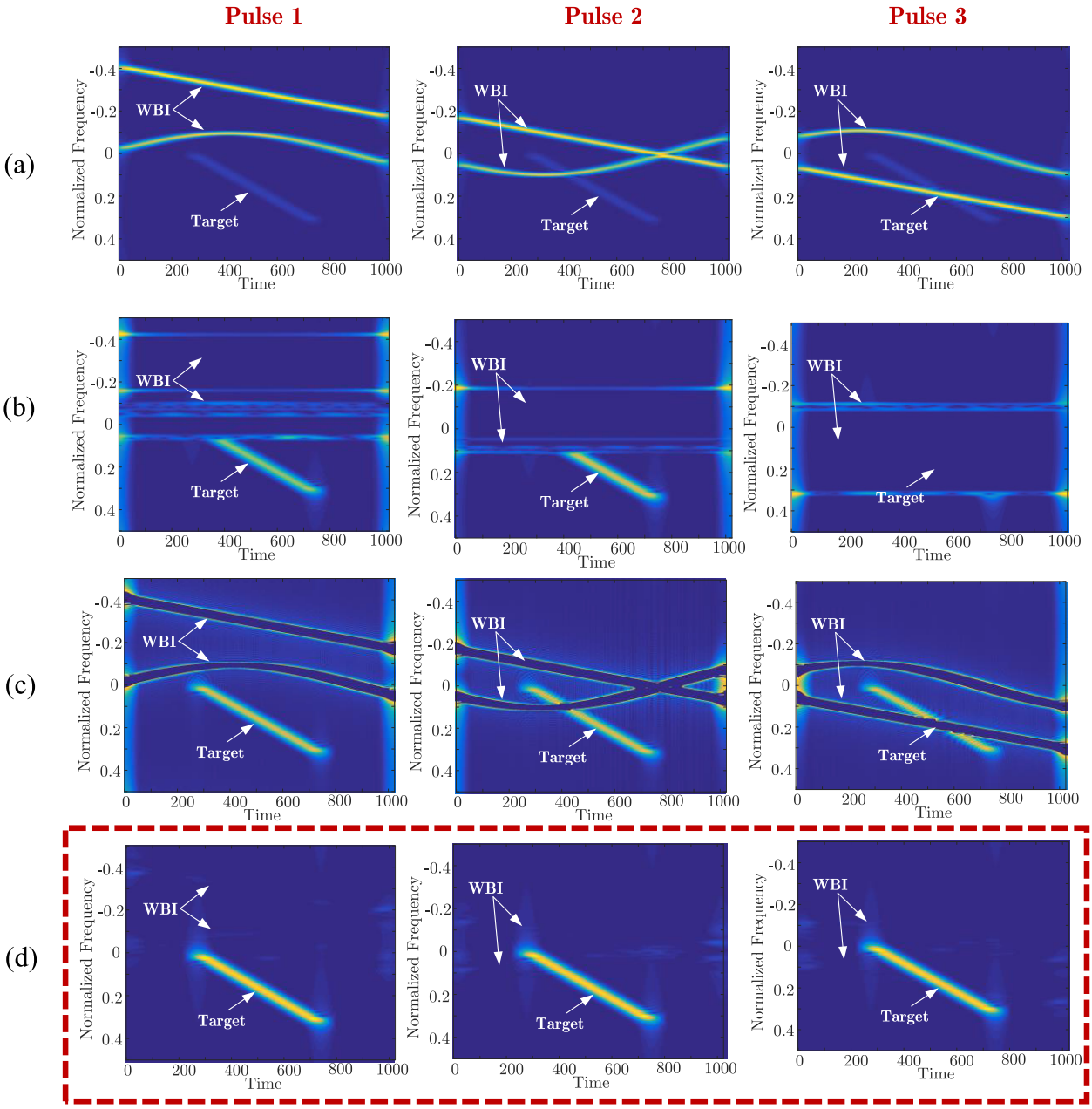


Fig. 7. Comparison between interference mitigation methods. (a) TF spectra of useful signal with WBI and TF spectra obtained by (b) frequency-notch filter, (c) TF-notch filter, and (d) proposed method.

TABLE II
EVALUATION METRICS OF THREE ALGORITHMS OF INTERFERENCE MITIGATION

Algorithm	Frequency-Notch filter[31]	TF-Notch filter[9]	RPCA-TFP-JDA
SDR (*dB*)	−2.5582	−7.7146	**-30.3747**

results after the frequency-notch filter, TF-notch filter, and the proposed method. Due to the loss of useful signals in the frequency domain and TF domain, imaging results present high sidelobe both in range and azimuth directions, as shown in Fig. 9(a) and (b), respectively. Meanwhile, a well-focused imaging result in both range and azimuth directions can be obtained after RPCA-TFP-JDA, as shown in Fig. 9(c). To further quantify the performance of the proposed method, peak sidelobe ratio (PSLR) and integrated sidelobe ratio (ISLR) [32], [33] are introduced to measure the imaging performance. To obtain the accurate values of the PSLR and ISLR, a noise-free scenario is considered. The PSLRs and ISLRs of imaging results with three mitigation methods are calculated and listed in Table III. It reveals that PSLR and ISLR obtained by RPCA-TFP-JDA are very close to the theoretical values, which indicates good preservation of the useful signal is guaranteed.

B. Choice of Parameters in RPCA-TFP-JDA

In this section, we discuss the choice of rank r and sparse degree ε in the proposed RPCA-TFP-JDA. Fig. 10 shows the

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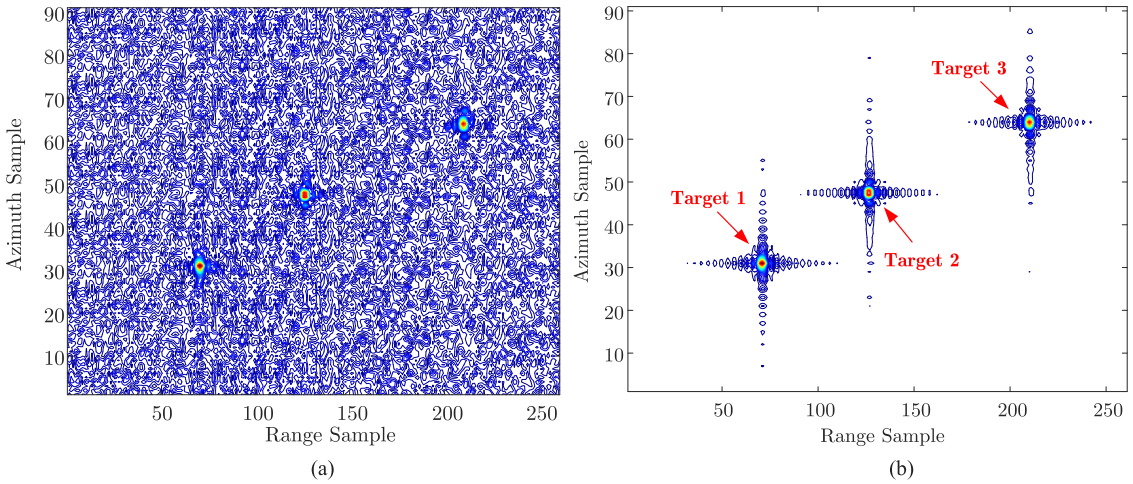


Fig. 8. Results of point-target imaging (a) with and (b) without WBIs.

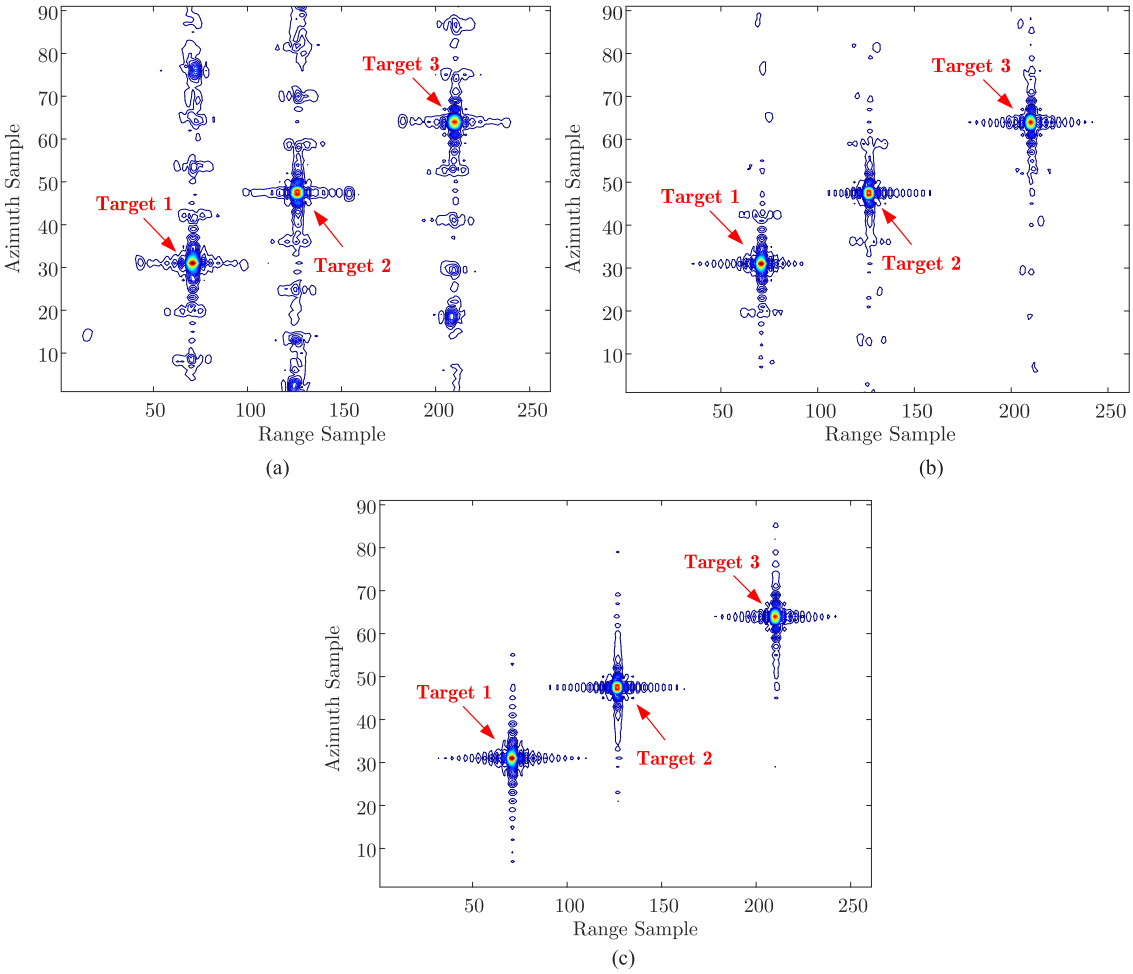


Fig. 9. Target imaging results after (a) frequency-notch filter, (b) TF-notch filter, and (c) RPCA-TFP-JDA.

variation of SDR values with rank. From Fig. 10, the SDR is the lowest of all when the rank r equals 1, while the SDR increases when the rank r is greater than 1. This phenomenon indicates that the useful signal in the TFP domain satisfies the low-rank property, which coincides with Remark I mentioned in Section III-C.

Fig. 11 discusses the selection of sparsity ε in the RPCA-TFP-JDA. From Fig. 11, the results of SDRs vary with sparse degrees, where ε changes from 800 000 to 1 700 000. The magnitude of sparse degree ε determines the number of nonzero elements in the sparse matrix. When the magnitude of sparsity is set too low, the sparse matrix covers only a small part of WBIs, causing residual interferences, while when the magnitude of sparsity is set too high, the useful signal may exist in the sparse matrix, resulting in the loss of the useful signals. To remove the interference cleanly and

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TABLE III
EVALUATION METRICS OF THREE ALGORITHMS OF INTERFERENCE MITIGATION FOR THREE TARGETS

Algorithm	Performance Measure		Target 1	Target 2	Target 3
Raw Data without WBI (Theoretical Values)	PSLR(*dB*)	Range	-13.51	-13.44	-13.39
		Azimuth	-10.66	-10.64	-10.63
	ISLR(*dB*)	Range	-12.13	-13.75	-13.54
		Azimuth	-8.99	-11.03	-10.62
Frequency-Notch Filter	PSLR(*dB*)	Range	-15.80	-15.65	-15.72
		Azimuth	-10.89	-12.69	-12.68
	ISLR(*dB*)	Range	-13.03	-12.45	-12.49
		Azimuth	-7.57	-9.05	-9.35
TF-Notch Filter	PSLR(*dB*)	Range	-14.81	-14.94	-15.00
		Azimuth	-11.92	-12.52	-11.59
	ISLR(*dB*)	Range	-12.42	-13.63	-13.67
		Azimuth	-8.80	-10.77	-10.69
Proposed Method	**PSLR(*dB*)**	**Range**	**-13.48**	**-13.44**	**-13.40**
		Azimuth	**-10.67**	**-10.65**	**-10.63**
	ISLR(*dB*)	**Range**	**-12.11**	**-13.75**	**-13.54**
		Azimuth	**-8.99**	**-11.04**	**-10.62**

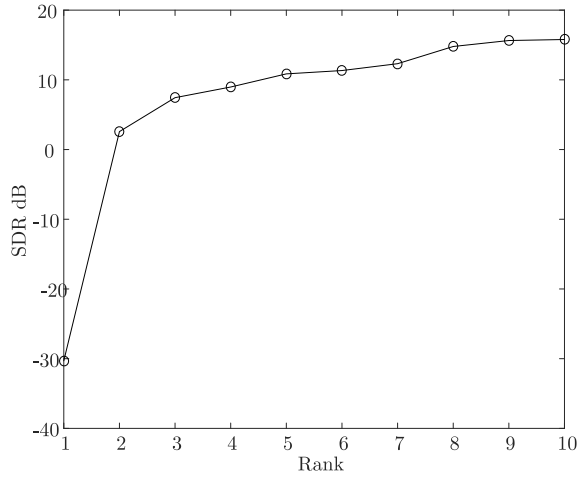


Fig. 10. SDR varying with rank.

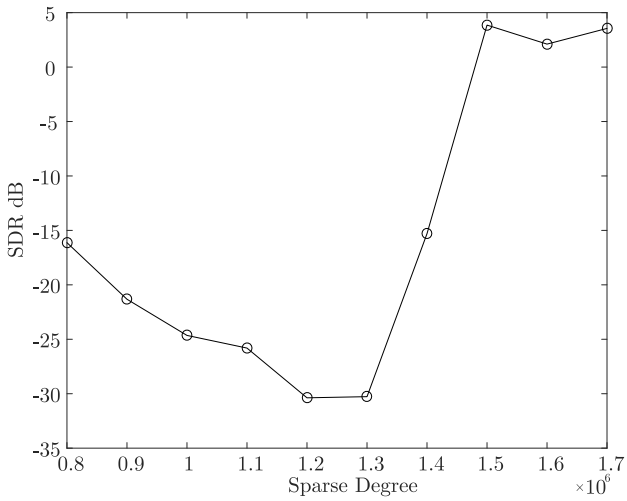


Fig. 11. SDR varying with sparse degree.

preserve the useful signal as much as possible, the optimal sparsity should be approximately equal to the number of points that WBI occupies in the TFP domain. Combined with the TF interference detector, the magnitude of sparsity can be obtained by

$$\varepsilon = \sum_{q}^{Q}\sum_{n}^{N}\sum_{k}^{M} \max\left\{\text{sign}\left(\log\left|\text{STFT}_q(n,k)\right| - \text{T}_h\right), 0\right\}. \quad (22)$$

According to Fig. 11, the SDR reaches the minimum when the sparsity is approximately equal to 1.2×10^6 (here, $\alpha = 1.8$).

C. Computational Complexity

The computational complexity of the frequency-notch filter, TF-notch filter, and the approach proposed in this article is listed in Table IV. For the proposed method, there are three main steps: STFT, RPCA, and ISTFT. First, the SAR echoes in adjacent pulses are transformed into the TF domain by the STFT operation, and its computational complexity is approximately equal to $O(Q\hat{N}M\log_2 M)$. Then, the RPCA is utilized to separate the TFP matrix of size $Q \times M\hat{N}$ into a low-rank matrix and a sparse matrix with the computational complexity of $O(r^2(Q + 3M\hat{N} + 4r) + (4\zeta + 4)QM\hat{N}r)$. After that, the WBIs are reconstructed by the ISTFT operation, whose computational complexity is $O(Q\hat{N}M\log_2 M)$.

D. Result of Measured Data

In this section, real measured data contaminated by serious interferences are used. To evaluate the image quality,

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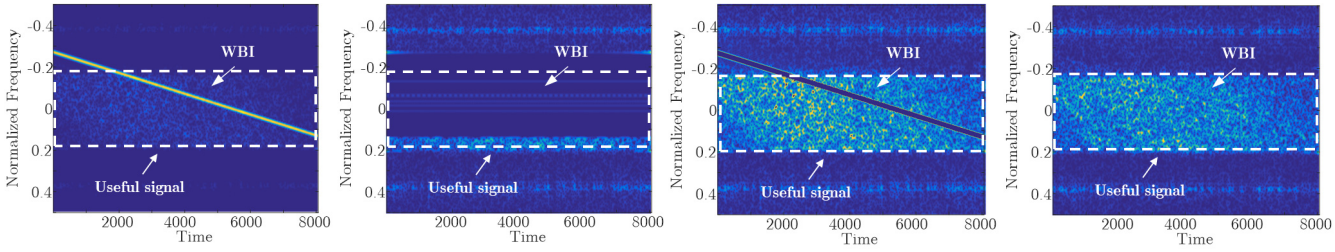


Fig. 12. Comparison between interference mitigation methods. (a) TF spectrum of measured data with CM-WBI and TF spectra obtained by (b) frequency-notch filter, (c) TF-notch filter, and (d) proposed method.

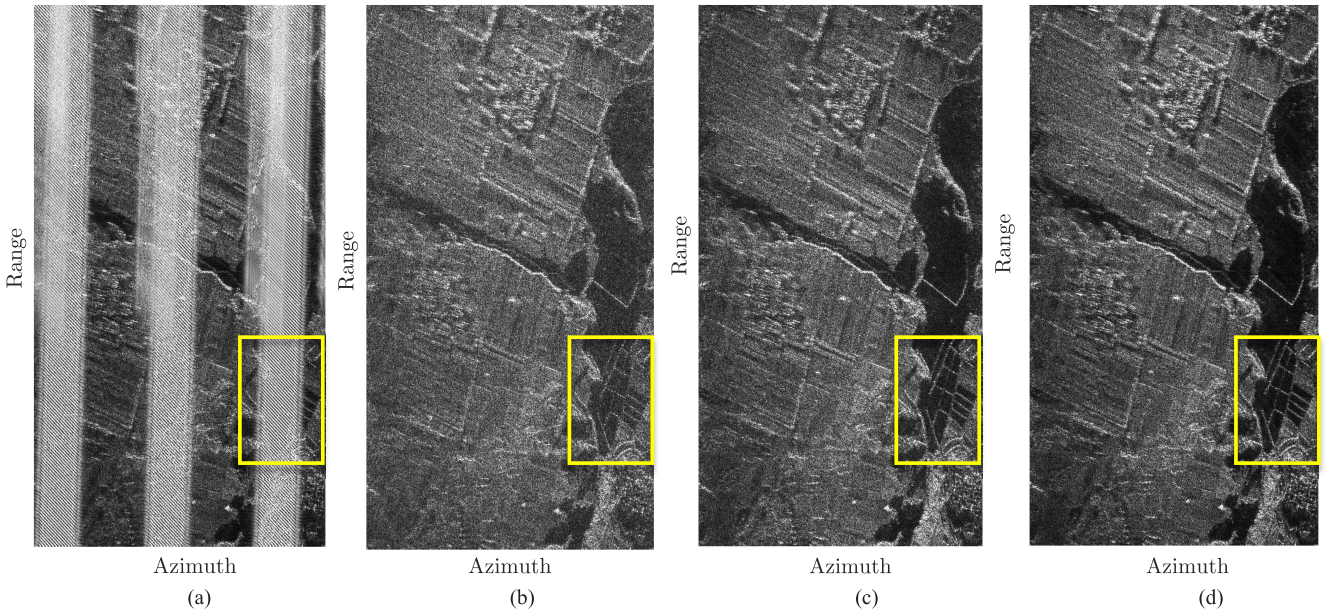


Fig. 13. Comparison of interference mitigation results. (a) SAR image with interference. (b) Frequency-notch filter. (c) TF-notch filter. (d) Proposed method.

TABLE IV
COMPUTATIONAL COMPLEXITY OF THREE ALGORITHMS

Algorithm	Main Step	Computational Cost
Frequency-Notch filter	FFT	$O(QN\log_2 N)$
	IFFT	$O(QN\log_2 N)$
TF-Notch filter	STFT	$O(Q\hat{N}M\log_2 M)$
	ISTFT	$O(Q\hat{N}M\log_2 M)$
RPCA-TFP-JDA	STFT	$O(Q\hat{N}M\log_2 M)$
	RPCA[18]	$O\left(r^2\left(Q+3M\hat{N}+4r\right)+\left(4\zeta+4\right)QM\hat{N}r\right)$
	ISTFT	$O(Q\hat{N}M\log_2 M)$

Note: Q is the total pulse number of received echo, N and M denote the length of signal and STFT window, respectively, $M\times\hat{N}$ is the size of the STFT matrix, where $\hat{N}=N-M+1$, and r denotes rank number. Besides, $\zeta \geqslant 0$ and the power scheme modification error of GoDec algorithm will reduce by increasing integer ζ.

we introduce the output SNR in the image domain, defined by [16]

$$\mathrm{SNR}=10\log_{10}\left(\frac{\sum_{i=1}^{N_1}|y_i|^2/N_1}{\sum_{j=1}^{N_2}|\hat{y}_j|^2/N_2}\right) \tag{23}$$

where y_i represents the ith prominent scatterer, $\hat{y}_j$ denotes the jth pixel of the surrounding region, and N_1 and N_2 denote the number of prominent scatterer pixels and noise pixels, respectively. From (23), it should be noted that the greater the SNR results in a clearer SAR image.

1) Result of CM-WBI Mitigation: The data set used was collected by the airborne SAR system. The TF spectra of real SAR data with simulated CM-WBI are shown in Fig. 12(a). Clearly, the TF spectrum of WBI occupies many frequency bins. Fig. 12(b) shows the TF spectrum of the signal with WBI after the frequency-notch filter. It is clearly shown that there is a great loss of useful signal. Compared with Fig. 12(b), Fig. 12(c) shows the result obtained by the TF-notch filter method with much less loss of useful signal. Fig. 12(d) shows the TF spectrum after RPCA-TFP-JDA. It is obvious from Fig. 12(d) that the proposed method can mitigate the WBI clearly and preserve the useful signal well.

In addition, SAR image results after interference mitigated by these three methods are presented in Fig. 13. The WBI-polluted SAR image is presented in Fig. 13(a), where the horizontal and the vertical axes represent the azimuth and range directions, respectively. From the figure, most portion of the scene is submerged in strong interferences, which hinders the interpretation of the interest scene or target. Therefore, WBI mitigation is necessary to obtain a well-focused SAR image. Fig. 13(b) and (c) presents the imaging results after the frequency-notch filter and TF-notch filter methods, respectively. Due to the loss of useful signals, the SAR imaging

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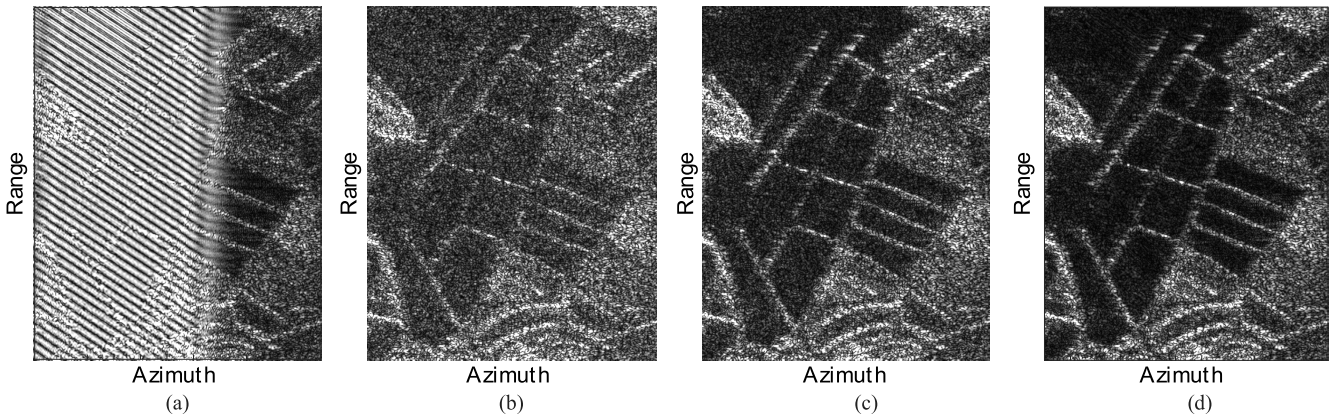


Fig. 14. Enlarged area of the yellow solid rectangle in Fig. 13. (a) SAR image with interference. (b) Frequency-notch filter. (c) TF-notch filter. (d) Proposed method.

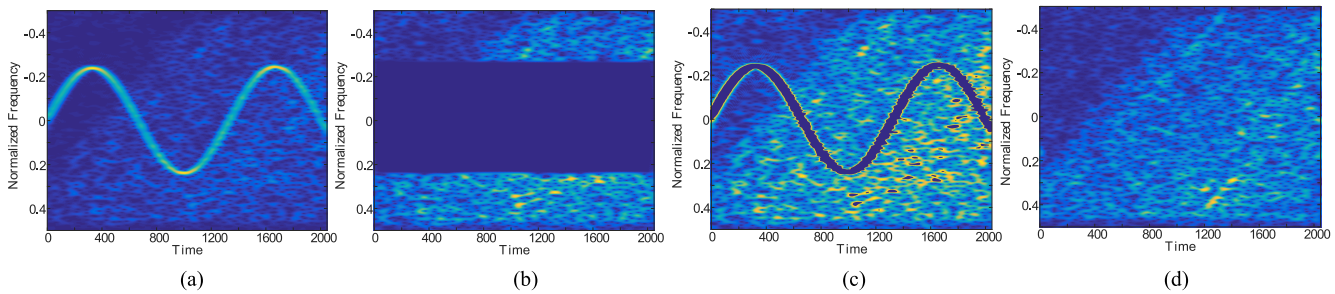


Fig. 15. (a) TF spectrum of measured data with interference. TF spectra obtained by (b) frequency-notch filter, (c) TF-notch filter, and (d) RPCA-TFP-JDA.

TABLE V
EVALUATION METRICS OF THREE METHODS FOR CM-WBI MITIGATION

Algorithm	Frequency-Notch filter[31]	TF-Notch filter[9]	RPCA-TFP-JDA
SNR (*dB*)	16.00	22.24	**25.16**

TABLE VI
EVALUATION METRICS OF THREE METHODS FOR SM-WBI MITIGATION

Algorithm	Frequency-Notch filter[31]	TF-Notch filter[9]	RPCA-TFP-JDA
SNR (*dB*)	24.4169	37.4742	**39.9379**

results are contaminated by noise. Fig. 13(d) shows the SAR image after RPCA-TFP-JDA. From Fig. 13(d), interferences are removed completely with a small loss of the useful signals, and thus, a clear SAR image, including village, mountain, river, and fields, is obtained. For a better comparison of interference mitigation performance, the enlarged details of rectangles in Fig. 13 are shown in Fig. 14.

The results of SNR for these three interference suppression algorithms are listed in Table V, indicating the superiority of the proposed method over the others. In conclusion, an RPCA-TFP-JDA-based interference algorithm can effectively suppress CM-WBI with the useful signal being kept well.

2) Result of SM-WBI Mitigation: The data used in this section are collected by another airborne SAR system. Fig. 15(a) shows the TF plots with the simulated SM-WBI, where most portion of the useful data is covered due to the existence of SM-WBI. The TF spectra obtained by three different mitigation methods are shown in Fig. 15. Since the interference and useful signal are highly overlapped in the frequency domain, a large amount of useful signal will be lost by the frequency-notch filter method, as shown in Fig. 15(b). The TF-notch filter can mitigate interferences effectively, and however, the useful signal, which is in the same TF domain with inferences, may also be suppressed, as shown in Fig. 15(c). Fig. 15(d) shows the result obtained by the proposed method. It is obvious that the interferences are suppressed clearly with little useful signal loss.

TABLE VII
TYPICAL PARAMETER VALUES IN AIRBORNE SAR SCENARIOS

Parameters	Typical Values
Vertical Distance (R_B)	20 *km*
Bandwidth (B)	100 *MHz*
Pulse Repetition Time(PRT)	1 *ms*
Velocity of Platform (V)	100 *m/s*
Pulse Duration (T_p)	10 *us*
Sampling Frequency (f_s)	200 *MHz*

Fig. 16 shows the SAR images obtained by three interference mitigation methods. Fig. 16(a) presents the SAR image with simulated SM-WBI, where the interferences overshadow

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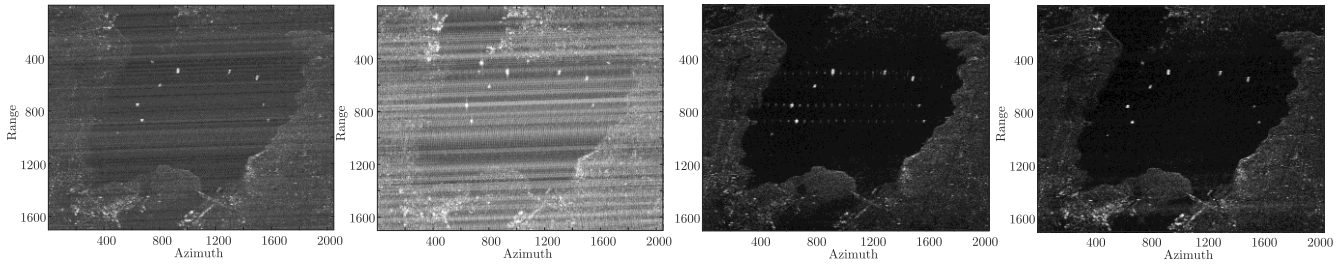


Fig. 16. Imaging results of WBI suppression. (a) SAR image contaminated by WBI. Imaging results obtained by (b) frequency-notch-filter, (c) TF-notch-filter, and (d) RPCA-TFP-JDA.

the important features, such as ship and field. Fig. 16(b) presents the SAR image after the frequency-notch filter. Because a significant portion of the useful signal is removed at each pulse, most bright stripes appear in the SAR image, and the SNR of the imagery is severely degraded. Fig. 16(c) shows the result obtained by the TF-notch filter. Compared with Fig. 16(b), the SAR image has more clear contours. However, the ghost targets appear in the SAR image due to discontinuities in the TF domain. Fig. 16(d) shows the SAR image obtained by the proposed method. It is observed that the image is well focused and interferences have been mitigated effectively. The SNRs for three mitigation methods are listed in Table VI, which indicates that the proposed method outperforms the frequency-notch filter and TF-notch filter in WBI mitigation. To sum up, the algorithm proposed in this article can effectively suppress WBIs with the useful signal being kept well.

V. Conclusion

The existence of WBI seriously decreases the SAR imaging quality. It is difficult to mitigate the WBI since it shares the same regions with useful signals in the time, frequency, and even TF domains. Traditional filtering-based WBI mitigation methods only exploit the 1-D or 2-D feature between useful signal and WBI, ignoring the feature difference between adjacent pulses.

In this article, an RPCA-TFP-JDA is proposed for time-varying WBI mitigation. First, as the TF amplitude of SAR echo follows the Weibull distribution, the log t-CFAR detector is selected for accurate WBI identification in each time slice of the TF domain. Then, we exploit the TFP joint feature difference between the useful signal and WBI. The useful signal in adjacent pulses has low-rank property, while WBI is treated as the sparse part. According to these properties, the useful signal and WBI in the TFP domain can be decomposed by the proposed method. Finally, the WBI is reconstructed and subtracted from the original signal to achieve interference mitigation. Compared with the traditional filtering-based methods, the effectiveness and superior performance of the proposed method are verified in both simulated and measured datasets. It should be noted that this algorithm is not suitable for NBI mitigation since the NBIs have the low-rank characteristic as the useful signal. Future work will introduce the tensor techniques for both NBI and WBI mitigation in the TFP domain.

Appendix A

This appendix provides a discussion on low-rank and sparse characteristics of useful signals and interference in the TFP domain.

Remark I: The TF spectra of useful signals in adjacent pulses are located in the same time and frequency bins, and thus, we can conclude that useful signals in the TFP domain satisfy the low-rank property.

The core issue is to demonstrate that the instantaneous frequency (IF) difference of useful signal between the qth and the $(q+p)$th pulses is much smaller than the sampling frequency gap. Assume that the SAR system transmits a linear-frequency-modulated signal (i.e., CM signal) in each pulse, denoted by

$$s_t(\hat{t}) = \mathrm{rect}\left(\frac{\hat{t}}{\mathrm{T}_p}\right)\exp\left(j\pi\gamma\hat{t}^2\right)\exp\left(j2\pi f_c\hat{t}\right) \tag{A1}$$

where $\mathrm{rect}(\hat{t}/\mathrm{T}_p)$ is a rectangle window function, $\hat{t}$ is the fast time, f_c denotes the carrier frequency, $\gamma = B/\mathrm{T}_p$ represents the chirp rate, and B and T_p denote the bandwidth and time width of the transmitted signal, respectively.

Assume that the ground scatterer is stationary, the received signal of the scatterer (i.e. useful signal) in the qth pulse is defined by

$$s_{r,q}(\hat{t}) = A_{r,q}\mathrm{rect}\left(\frac{\hat{t}-\tau_q}{\mathrm{T}_p}\right)\exp\left[j\pi\gamma\left(\hat{t}-\tau_q\right)^2\right]\exp\left(-j2\pi f_c\tau_q\right) \tag{A2}$$

where $A_{r,q}$ is the amplitude of the received signal in the qth pulse and τ_q is the delay time in the qth pulse, which is defined by

$$\tau_q = \frac{2R(V,t_q)}{c} = \frac{2}{c}\sqrt{\left(Vt_q - x_P\right)^2 + y_P^2 + (H-z_P)^2} \tag{A3}$$

where V and c present the speed of the SAR system and light, respectively, $t_q = q\cdot\mathrm{PRT}$ is the slow time, and PRT is the pulse repetition time. For convenience, we consider $x_P = 0$ and $z_P = 0$. Equation (A3) is rewritten as

$$\tau_q = \frac{2}{c}\sqrt{\left(Vt_q\right)^2 + y_P^2 + H^2} = \frac{2}{c}\sqrt{R_B^2 + \left(Vt_q\right)^2} \tag{A4}$$

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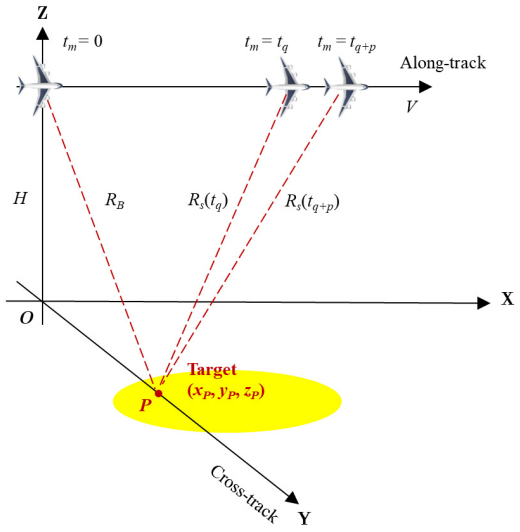


Fig. 17. Geometry of the side-looking SAR system.

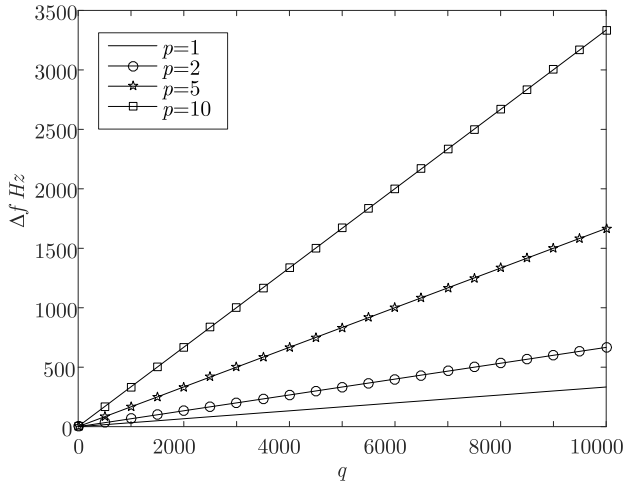


Fig. 18. IF difference Δf varying with parameters q and p.

where R_B is the vertical distance (or the closest slant range) from the radar platform flight path to point P, as shown in Fig. 17.

Due to $Vt_q \ll R_B$, (A4) is rewritten as follows:

$$\tau_q \approx \frac{2}{c}\left(R_B + \frac{1}{2}\frac{V^2}{R_B}t_q^2\right). \tag{A5}$$

The IF of the received signal in the qth pulse is

$$\begin{aligned} f_q &= \frac{1}{2\pi}\frac{d\left[\pi\gamma\left(\hat{t}-\tau_q\right)^2 - 2\pi f_c \tau_q\right]}{d\hat{t}} = \gamma\hat{t} - \gamma\tau_q \\ &= \left(\gamma\hat{t} - \gamma\frac{2R_B}{c}\right) - \gamma\frac{V^2}{cR_B}t_q^2 \end{aligned} \tag{A6}$$

and the IF difference of useful signal between the qth and the $(q+p)$th pulses is

$$\begin{aligned} \Delta f &= \left|f_q - f_{q+p}\right| = \gamma\frac{V^2}{cR_B}\left(t_{q+p}^2 - t_q^2\right) \\ &= \gamma\frac{V^2\mathrm{PRT}^2}{cR_B}(2q+p)p. \end{aligned} \tag{A7}$$

If (A7) satisfies the following condition, i.e.,

$$\Delta f \ll \frac{f_s}{N_{\mathrm{FFT}}} \tag{A8}$$

where f_s is the sampling frequency and N_{FFT} is the number of Fourier transform points in the STFT operation, we can conclude that these useful signals are located in the same frequency bins. Typical parameter values for the SAR system are listed in Table VII. The sampling frequency gap is about 391 kHz when $f_s = 200$ MHz and $N_{\mathrm{FFT}} = 512$. Fig. 18 shows the IF difference Δf varying with parameters q and p. It can be seen from Fig. 18 that the maximum value of IF difference is 3335 Hz at $p = 10$ and $q = 10\,000$ (i.e., $t_q = 10$ s), which is much smaller than the sampling frequency gap, satisfying (A8). Thus, we have

$$\mathbf{STFT_{U,q}} \approx \mathbf{STFT_{U,q+1}} \approx \cdots \approx \mathbf{STFT_{U,q+p}} \tag{A9}$$

which proves that the STFT matrices of useful signals in adjacent pulses satisfy the low-rank characteristic. For the moving target, if the IF difference between the qth and the $(q+p)$th pulses satisfies (A8), the moving target can be considered to have a low-rank property.

Remark II: The TF spectra of WBIs are sparsely distributed in adjacent pulses with different locations, and thus, CM-WBI and SM-WBI have sparse characteristics.

Different from the transmitted signal, interferences are noncooperative and their parameters (such as frequency, chirp rate, and modulation index) may vary significantly between adjacent pulses. For example, assuming that WBIs in adjacent pulses are continuous, the frequencies of CM-WBI and SM-WBI in the qth pulse can be written as

$$\begin{aligned} f_{\mathrm{CM}-WBI,q} &= f_{\mathrm{CM}} + \kappa(n + qN) \\ &= f_{\mathrm{CM},q} + \kappa n \\ f_{\mathrm{SM}-WBI,q} &= f_{\mathrm{SM}}\beta_{\mathrm{SM}}\cos[2\pi f_{\mathrm{SM}}(n+qN)] \\ &= f_{\mathrm{SM}}\beta_{\mathrm{SM}}\cos\left(2\pi f_{\mathrm{SM}}n + \varphi_{\mathrm{SM},q}\right) \end{aligned} \tag{A10}$$

where $f_{\mathrm{CM},q} = f_{\mathrm{CM}} + \kappa qN$ and $\varphi_{\mathrm{SM},q} = 2\pi f_{\mathrm{SM}}\ qN$ represent the initial frequency of CM-WBI and initial phase of SM-WBI in qth pulse, respectively. In (A10), it is obvious that the initial frequencies and phases of CM-WBI and SM-WBI in adjacent pulses are completely different, as shown in Fig. 6 marked by ellipses. Thus, we can obtain

$$\mathbf{STFT_{I,q}} \neq \mathbf{STFT_{I,q+1}} \neq \cdots \neq \mathbf{STFT_{I,q+p}} \tag{A11}$$

which proves that the STFT matrices of CM-WBI and SM-WBI in adjacent pulses do not have low-rank characteristics, while the TF spectra of CM-WBI and SM-WBI are sparsely distributed in adjacent pulses, as shown in Fig. 6; thus, we conclude that they have sparse characteristics.

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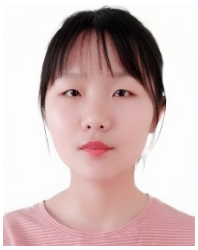
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